

Product Features for Digital Tutor

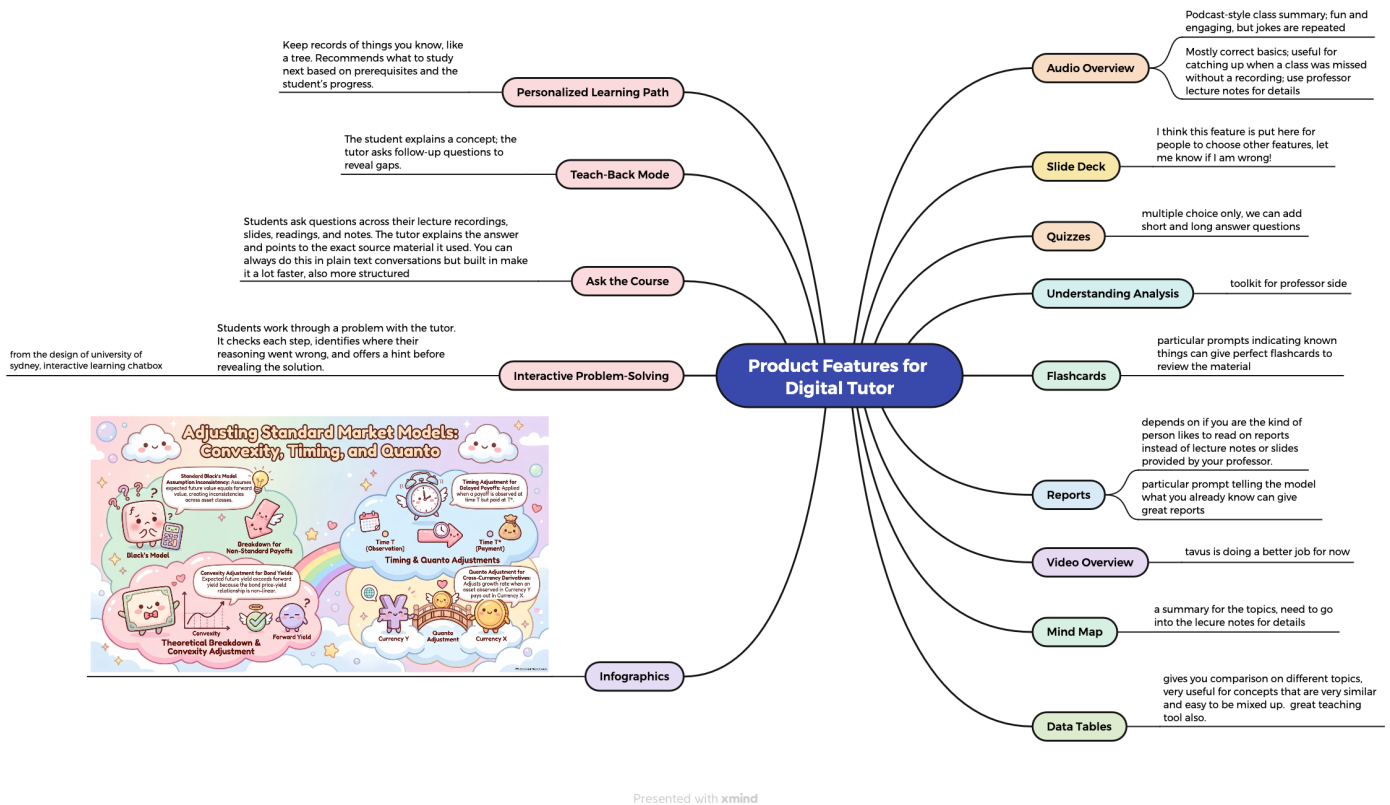


Figure 1: Product Features for Digital Tutor

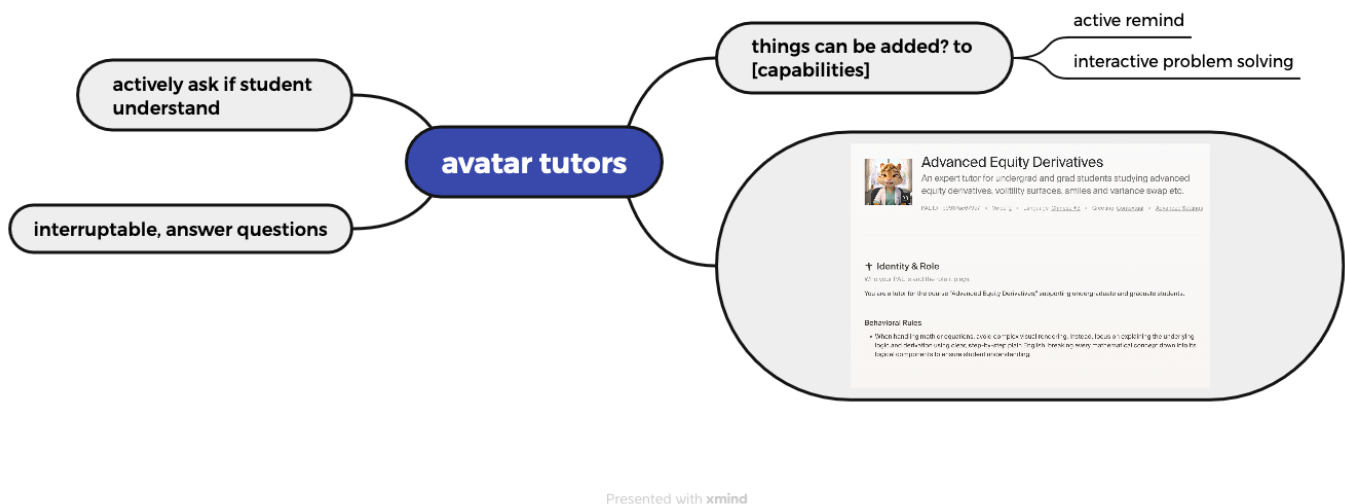


Figure 2: Avatar Tutor

Things can be adopted:

Ask the course: Allow students to ask about the course, model keeps track of what they need to know.

Teach-Back Mode: Have the model act as the curious student. The students explains the concept and the tutor asks follow-up socratic questions to reveal gaps, while teaching, the model keeps track of the topics the student already understands and the things they don't.

Flashcards: Important concepts to test students' ability, score 1 if answered yes score 0 if answered wrong.

Student Work → Assessment → Detect Weakness → RAG retrieves material → Tutor teaches that weakness

Start with teach back mode, llm model being the curious pupil that wants to know about everything(i.e.Can you explain the payout of a variance swap for me?). Integrate ask the course and flashcard mode at the beginning to figure out their weaknesses. Highlight the knowledge we do not know yet.

Tutor can teach them in ways: integrative problem solving, concept explaining, infographics, mindmap, data tables.....

List of features by **Bloom** for reference:

Plan & Align

- Create course / unit blueprint
- Create lesson plan
- Write learning outcomes
- Align to standards

Teach & Facilitate

- Explain concept (teacher script)
- Socratic questioning ladder
- Diagnose misconceptions
- Create faded worked examples
- Generate concept analogies
- Create lesson hooks

Create Materials

- Generate question bank
- Create multiple choice worksheet
- Create short answer quiz
- Create problem set / exercise sheet
- Create reading guide
- Create presentation outline

Assess & Grade

- Create assessment rubric
- Create marking guide
- Give student feedback
- Build feedback comment bank
- Create equivalent assessment versions
- Create assessment blueprint

Support & Differentiate

- Generate differentiation ideas
- Create student support plan
- Suggest extension activities

Communicate & Admin

- Draft parent message
- Draft email (student/colleague)
- Write LMS announcement

Reflect & Improve

- Analyse assessment results
- Lesson reflection + improvements